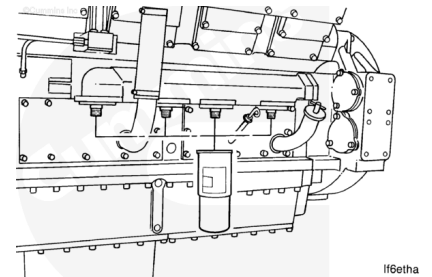


Trainings ▾

Remove

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



⚠ CAUTION ⚠

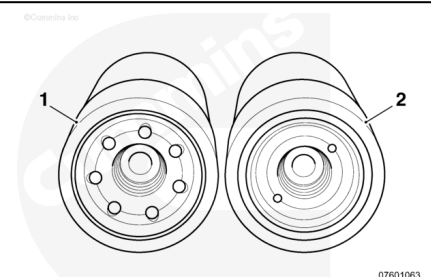
Use caution when draining oil or replacing filters that the oil is not spilled into the bilge area. The oil and oil filters must be disposed of in accordance with local environmental regulations.

K50 engines have five full-flow oil filters.

Use an oil filter wrench, Part Number 3400157, or equivalent.
Remove the full-flow filters.

Discard the filters if they are **not** needed for a failure analysis.

The external appearance of the full flow (1) and the bypass (2) filters is the same. The accompanying graphic shows the difference between the two filters.

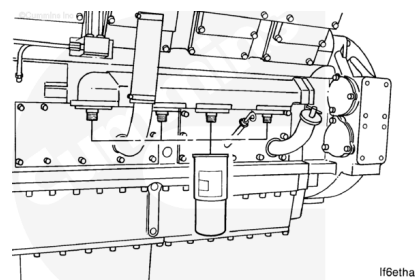


Install

Note : Most K50 engines require five full-flow oil filters.

Use clean engine oil. Fill the filter with oil. Lubricate the seals and the portion of the filter adapter that does **not** contain threads.

Install the filters. Refer to the OEM service manual.



If6etha

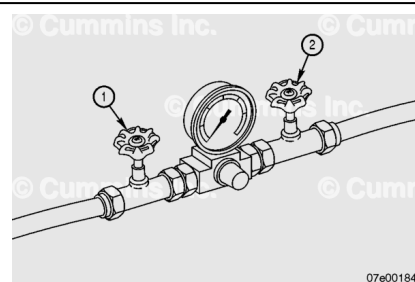
Pressure Differential Test

Spin-on Type

This test will indicate oil pressure before and after the oil flows through the filters.

Use a differential pressure gauge or one pressure gauge with two oil hoses and two valves (1 and 2) to eliminate gauge error. Use a gauge with a minimum pressure capacity of 1400 kPa [200 psi].

Be sure the engine is switched OFF.



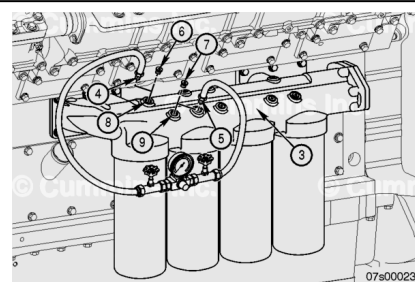
07e00184

Remove the plug (7) from the outlet end (9) of the oil filter head (3) and install the end (5) of the valve assembly in the port of the filter head from where the plug (7) has been removed.

The reading will indicate the oil pressure after the oil flows through the filters.

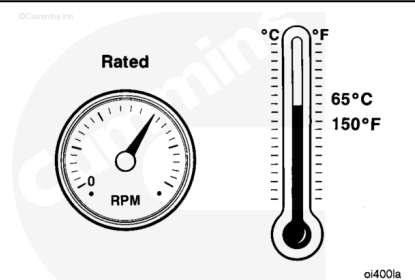
Remove the plug (6) from the inlet end (8) of the filter head (3) and install the end (4) of the valve assembly oil hose into where the second plug (6) has been removed.

This reading will indicate the oil pressure before the oil flows through the filters.



07s00023

Operate the engine at rated rpm until the engine oil reaches its normal operating temperature. Continue running the engine at rated rpm for the duration of the pressure test.

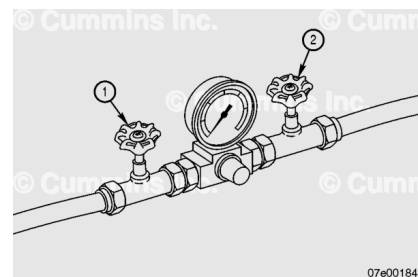


o/400la

Close valve one (1) and open valve two (2) to read the oil pressure before the filters.

Close valve two (2) and open valve one (1) to read the oil pressure after the filters.

If the difference in pressure is more than the maximum allowable value, an oil filter with excessive restriction is being used. Replace the filters and check the pressure differential again.



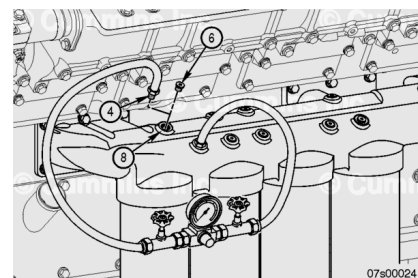
Measurements			
		kpa	psi
Maximum Oil Filter Pressure Differential	New/clean oil filters	83	12
	Used oil filters	345	50

Switch the engine OFF.

Remove the hose (4) from the inlet end of filter head (3) and install the plug (6).

Tighten the plug.

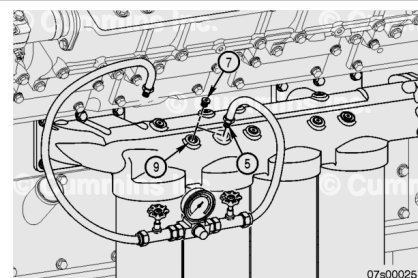
Torque Value: 15 n•m [133 in-lb]



Remove the hose end (5) from outlet end of filter head (9) and install the plug (7).

Tighten the plug.

Torque Value: 15 n•m [133 in-lb]



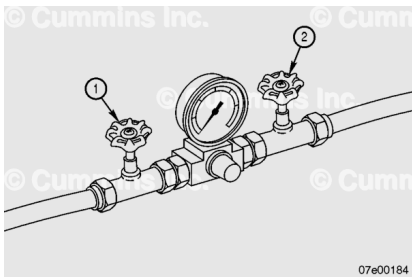
Eliminator™

This test will indicate oil pressure before and after the oil flows through the Eliminator™.

Use a differential pressure gauge or one pressure gauge with two oil hoses and two valves (1 and 2) to eliminate gauge error. Use a gauge with a minimum pressure capacity of 1400

kPa [200 psi].

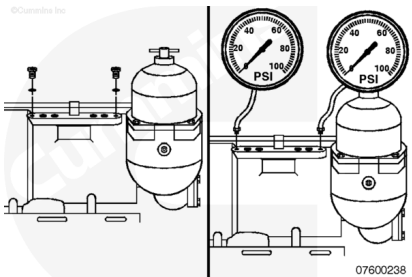
Be sure the engine is switched OFF.



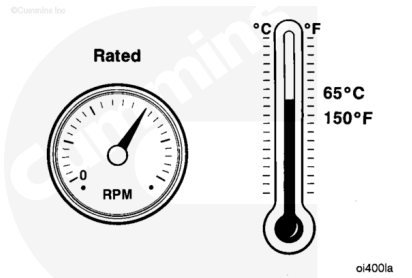
Remove the two plugs from the filter housing.

Install one of the valve assemblies in the port of the filter nearest to the centrifuge from where the plug has been removed.

Install the other valve assembly in the port of the filter furthest away from the centrifuge from where the other plug has been removed.



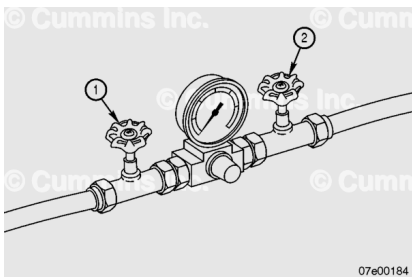
Operate the engine at rated rpm until the engine oil reaches its normal operating temperature. Continue running the engine at rated rpm for the duration of the pressure test.



Close valve one (1), open valve two (2), and record the pressure value before the Eliminator™.

Close valve two (2), open valve one (1), and record the pressure value after the Eliminator™.

If the difference in pressure is more than the maximum allowable value, inspect and clean the Eliminator™ screens before checking the oil pressure differential again.



Measurements

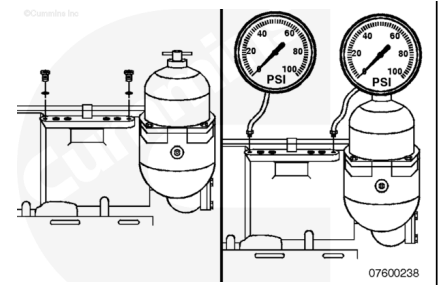
	kpa	psi
Maximum Oil Pressure Differential	60	9

Switch the engine OFF.

Remove the valve assemblies from the Eliminator™ and install the two plugs.

Tighten the plugs.

Torque Value: 10 n•m [89 in-lb]



Last Modified: 25-May-2021
